

Shubhankar Tiwari

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building enterprise backend systems at scale, transitioning into ML/AI engineering with hands-on development of LLM evaluation platforms, model governance pipelines, and AI reliability systems. Incoming M.S. in Artificial Intelligence at Columbia University. Combines proven production engineering discipline with applied AI systems depth, targeting ML/AI Engineer roles at large tech and fintech companies.

EDUCATION

Columbia Engineering

M.S. in Artificial Intelligence.

New York, NY
Dec 2027 (Expected)

SRM Institute of Science and Technology

B.Tech, Computer Science and Engineering. GPA: 94.4%.

2022

EXPERIENCE

Software Engineer

Bank of America

Apr 2022 – Jul 2026
Hyderabad, India

- Develop and maintain enterprise backend services for CashPro Search, serving 550,000+ corporate users across 40,000 companies with 18 million+ searches since launch.
- Own OpenShift deployments, CI/CD configuration, release validation, and production stabilization across 17+ release cycles.
- Serve as Subject Matter Expert for CashPro Search, providing technical guidance, production troubleshooting, and knowledge transfer across engineering, QA, and release teams.
- Lead modernization initiatives including JDK 17, Spring 6, containerization, cloud enablement, and SonarQube-driven quality improvements across 13 CashPro services.
- Diagnose and resolve 50+ production issues involving SSL certificates, database connectivity, Drools dependencies, deployment failures, and privacy-related defects.

Product Development Trainee

HighRadius Technologies

Aug 2021 – Mar 2022

- Built backend modules for Order-to-Cash and Treasury Systems using Java, JavaScript, and Python, contributing to Agile sprint delivery and enterprise API integrations.

PROJECTS

AI Reliability Platform for LLM Evaluation

2025 – Present

- Built an LLM reliability platform using entropy-aware routing, adaptive regeneration policies, and behavioral workflows.
- Engineered an adversarial noise injector and async pipeline to evaluate multi-model stability across Llama, Mistral, and Qwen; achieved 100% harmful-output catch rate on 43 adversarial prompts across 9 blind holdout rounds.
- Deployed a 9-category guardrail classifier with escalation logic; recorded 0% leakage and 0% hallucination across 213 evals.
- Designed 55+ automated regression tests and CI-gated promotion workflows that blocked 7/7 unsafe model candidates and reduced deployment fragility 66% (25.0 to 8.5).

Drift-Aware Fraud Detection & Model Governance

2025 – Present

- Built a drift-aware ML governance pipeline using PSI drift detection, threshold-aware promotion policies, and loss-based scoring.
- Implemented behavioral regression detection and CI-gated promotion workflows to block misleading models from deployment.

AWARDS & HONORS

- Global Recognition Awards (Gold, Silver, Bronze) at Bank of America for backend delivery, cloud enablement, OpenShift migration, production stabilization, and release ownership across the CashPro platform.
- Kaggle Notebooks Expert: Personal Best #707; 10 Bronze Medals; Machine Learning, Deep Learning, and Computer Vision certified.

SKILLS

AI/ML: Python, PyTorch, Transformers, Scikit-learn, XGBoost

ML Systems: LLM Evaluation, AI Reliability, Model Governance, Behavioral Testing, Experimentation Frameworks, Model Validation

Backend: Java, Spring Boot, REST APIs, SQL, OpenShift, Jenkins, Ansible, Linux